Original Article

Geospatial Network Analysis of Maternal Healthcare Access and Travel Impedance in South 24 Parganas District, West Bengal

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Abstract
Background: Geographic disparities in access to emergency obstetric care contribute heavily to maternal mortality. Spatial barriers often delay life-saving clinical interventions. **Objectives:** This study aimed to model the travel network impedance from localized population centres to tertiary and secondary referral health facilities in South 24 Parganas, West Bengal. **Methodology:** A secondary spatial data analysis was conducted for South 24 Parganas district, bounded between 21°29′ N to 22°33′ N latitude and 88°3′ E to 89°5′ E longitude. Eight population nodes and five institutional health hubs were mapped. QGIS software v3.36 was used to calculate travel time matrices using existing road and waterway network vector layers. Bed counts and local population densities served as core attribute variables. **Results:** Calculated travel costs from population nodes to the closest facilities varied significantly across the geographic landscape. The minimum travel impedance recorded was 19.55 minutes; the maximum reached 125.15 minutes. Remote deltaic sectors demonstrated extreme transit times to higher-order health facilities. **Conclusion:** Severe spatial constraints restrict healthcare access for maternal populations within remote hubs of South 24 Parganas. Targeted infrastructure investment and strategic placement of emergency obstetric care facilities are crucial to bridge these geographic gaps.

Key Words: Geographic Information Systems, Health Services Accessibility, Maternal Health Services, Maternal Mortality, Travel Time

Key Message: Geospatial network modeling reveals critical gaps and extreme travel-time variation — up to 125 minutes — for maternal populations accessing secondary and tertiary referral health hubs in South 24 Parganas.

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I. Introduction

Maternal mortality remains a significant global public health concern[1]. Type II delay in reaching an appropriate medical facility is a major driver of preventable maternal deaths[2,3]. Spatial distance and poor transport infrastructure aggravate these barriers[4], a problem particularly acute in developing countries and rural sub-regions[5].

South 24 Parganas district, with an approximate maternal mortality ratio of 110 per one lakh live births, has unique and complex geographical terrain[6]. The deltaic region contains extensive networks of rivers, creeks, and waterways[7], and this estuarine terrain hinders continuous road infrastructure development. Consequently, pregnant women in remote pockets face immense physical challenges during medical emergencies, navigating fragmented roads and seasonal river transport to reach healthcare facilities[8].

Traditional administrative planning often overlooks real-world transit layouts[9], and straight-line distance measurements do not represent true travel impedance over complex terrain[10]. Geographic Information Systems (GIS) provide advanced tools to address this limitation[8]: network analysis techniques calculate precise travel times along actual transport vectors[10], incorporating physical barriers, road classes, and transport speeds[9], offering a more realistic assessment of maternal healthcare accessibility than straight-line models.

A detailed evaluation of the existing referral network is therefore necessary[8], potentially allowing health administrators to optimize resource distribution[5] and place emergency obstetric care facilities strategically[3]. This study was conceptualized to fill existing gaps in regional transport planning for maternal health, analyzing spatial network accessibility and travel-time impedance from population centres to secondary and tertiary healthcare hubs in South 24 Parganas district.

II. Methodology

Study design and area

This was a secondary data analysis using geospatial modeling techniques, conducted for South 24 Parganas district, West Bengal, bounded between 21°29′ N to 22°33′ N latitude and 88°3′ E to 89°5′ E longitude[6]. The district includes a mix of peri-urban plains and the riverine tracts of the Sundarbans delta[7].

Network nodes and hubs

The spatial network model incorporated specific geographic points to assess travel friction[9]. Eight population clusters were selected as origin nodes, representing major community settlements: Budge Budge, Sonarpur, Jaynagar, Mograhat, Mathurapur, Kulpi, Falta, and Basanti. Five institutional health facilities were designated as destination hubs: Diamond Harbor Medical College, Baruipur Sub-Divisional Hospital (SDH), Canning SDH, Sarisha Block Primary Health Centre (BPHC), and Padmahat BPHC.

Study variables and data sources

The primary non-spatial attribute was total bed count at each health facility hub[6]; local population numbers at each settlement node were included as a demand variable. Primary spatial layers consisted of road networks, navigable waterways, and health-hub point locations. Secondary spatial layers, transport networks, and demographic variables were sourced from official public-domain repositories and institutional health records.

Data analysis and spatial modeling

Spatial network analysis was executed using open-source QGIS software version 3.36 to compile the geographic database and run network simulations. A digital network graph was constructed from vector road and waterway layers, with travel speeds assigned to road categories per standard transport guidelines and waterway transit parameters added to represent ferry crossings in deltaic zones.

The Network Analysis toolbox was used to run an origin-destination cost matrix[9], calculating shortest travel paths along network vectors[10]. Accumulated travel time in minutes was computed from each of the eight origin nodes to each of the five destination hubs, accounting for land-water transitions and topological connectivity, and mapped to produce a Maternal Access Index. A secondary proximity filter was then applied to isolate and map only the single least-time-consuming path from each origin node to its nearest accessible facility.

Ethical considerations

This research was based entirely on secondary data layers containing no individual human participant identifiers; no clinical records or personal health information were accessed. This study therefore did not require formal ethical clearance or institutional review board permission.

III. Results

The spatial distribution of health facilities and travel-time categories across South 24 Parganas was evaluated systematically. The network analysis quantified travel impedance between all nodes and hubs (Table 1). The Maternal Access Index is divided into four tiers: green (20–50 min), light green (50–80 min), orange (80–110 min), and red (110–140 min).

Node	Hub 1	Hub 2	Hub 3	Hub 4	Hub 5
1. Budge Budge	33.83	76.17	125.15	40.75	79.87
2. Sonarpur	25.82	79.17	99.30	19.55	70.77
3. Jaynagar	30.67	93.64	109.15	28.46	80.65
4. Mograhat	75.87	70.16	119.46	82.79	83.29
5. Mathurapur	89.82	19.95	64.32	89.58	33.07
6. Kulpi	54.04	50.57	67.65	47.20	37.58
7. Falta	53.72	64.75	80.27	46.88	51.76
8. Basanti	120.84	77.61	35.19	114.00	69.74

Table 1: *Travel time impedance matrix from population nodes to health hubs, in minutes. Hub 1: Diamond Harbor Medical College; Hub 2: Baruipur SDH; Hub 3: Canning SDH; Hub 4: Sarisha BPHC; Hub 5: Padmahat BPHC.*

The lowest travel time recorded (19.55 min) occurs from Sonarpur to Sarisha BPHC (Hub 4); the highest (125.15 min) occurs from Budge Budge to Canning SDH (Hub 3). Basanti shows a low travel time of 35.19 min to Hub 3 but an extreme 120.84 min to Hub 1. Mograhat maintains a travel time above 70 minutes across all five destination hubs.

The time-optimized pathway analysis mapped actual physical transit routes along the road and waterway infrastructure. *Figure 2* Figure 2a describes the exhaustive network of all potential spatial paths; the time-optimized pathways in Figure 2b demonstrate that rapid healthcare access (under 29 minutes) remains heavily restricted to the western and northwestern perimeters, leaving communities adjacent to the southern deltaic networks isolated with significantly higher travel times to secondary or tertiary clinical infrastructure.

IV. Discussion

The main objective of this study was to analyze spatial network accessibility and travel-time impedance from population nodes to health hubs in South 24 Parganas. Findings showed wide variation in travel impedance across the district.

The estuarine geography of South 24 Parganas creates natural obstacles for land transport[7]. Localities like Basanti are separated from major urban centres by rivers and creeks, and reliance on ferry connections increases transit times significantly[6] — explaining the high travel cost of 120.84 minutes from Basanti to Diamond Harbor Medical College. Conversely, sectors with contiguous highway networks show much lower travel impedance: Sonarpur, for example, shows a short travel time of 19.55 minutes to Sarisha BPHC due to better road connectivity. Fragmented infrastructure directly compromises maternal health outcomes[5]: long transit times prolong the second delay during obstetric emergencies[2], increasing the risk of birth complications and maternal mortality[4]. Pockets facing travel times above 80 minutes require urgent attention, as these high-impedance zones leave pregnant women vulnerable when complications arise[8].

When the network is optimized to model transit strictly to the closest available facility, a different layer of spatial inequality emerges.

Even under this ideal-case scenario, the safety net is not uniform: while western border populations enjoy efficient access within 29 minutes, northern and sub-deltaic communities still experience localized infrastructure friction, with nearest-facility transit stretching up to 71 minutes. This implies that map proximity does not automatically equate to rapid clinical access when underlying road vectors are fragmented.

This study used actual road and waterway network paths rather than straight-line distances, providing a realistic model of travel impedance over difficult deltaic terrain. The dual-model approach — a comprehensive multi-destination matrix alongside a proximity-filtered nearest-hub analysis — enhances the accuracy and utility of the geographic model for real-world referral planning. However, the network analysis assumed static travel speeds and did not account for real-time traffic congestion or seasonal weather disruptions; monsoonal flooding often slows transit or disrupts ferry services entirely in the Sundarbans. The model also did not include private transport modes or emergency ambulance availability.

Localities with travel times exceeding 100 minutes need immediate decentralized care options, including specialized obstetric stabilization units at riverine transit points, improved multi-modal transport networks, and upgraded ferry services and bridges to reduce travel friction. Future research should incorporate seasonal variables — particularly monsoon-period transit variation — and examine how facility capacity affects patient choice, combining spatial travel costs with clinical quality data to further optimize maternal health planning.

V. Conclusion

This study highlights major geographic disparities in maternal healthcare access across South 24 Parganas. Network modeling revealed travel times to referral hubs ranging from under 20 minutes to over two hours, with the riverine landscape creating significant transit barriers for remote communities. These findings emphasize that straight-line distance models underestimate actual travel barriers. Reducing these spatial inequities requires targeted infrastructure investment and decentralized emergency obstetric facilities, essential to eliminating transit delays and lowering maternal mortality.

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